

SPEED CONTROLLER FOR AC AND DC MOTORS

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In manufacturing industries like plastic, textile, chemical and pharmaceuticals, speed control of AC and DC motors is important for efficient and consistent production. Depending upon the requirement and application, the motor may be huge or small. Various AC/DC drives are available for speed control of these motors. But one has to use separate drives to control AC and DC motors. Here is a circuit that can control the speed of both AC as well as DC motors.

Circuit description

Fig. 1 shows the circuit for speed control of AC and DC motors. It comprises regulator 7805, timer NE555, optocoupler MOC3021, SCR 2P4M and triac BT136 along with two 2N2222 transistors and a few discrete components.

To derive the power supply for the circuit, the 230V, 50Hz AC mains

is stepped down by transformer X1 to deliver secondary output of 12V, 2A.

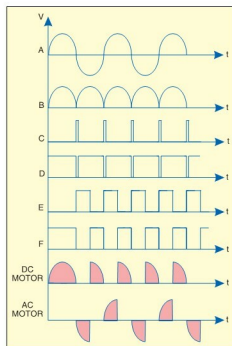


Fig. 2: Waveforms at various points of the circuit

The transformer output is rectified by full-wave bridge rectifier BR1, filtered by capacitor C1 and regulated by IC 7805 (IC1). Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and resistor R1 limits the current through LED1. Diode D2 couples the rectified output to the base of transistor T1 through the voltage divider formed by resistors R2 and R3.

The collector of transistor T1 is connected to the base of transistor T2 through resistor R4. The signal at the collector of T2 triggers pin 2 of timer IC2. Timer IC2 is configured as a monostable multivibrator and its time period is determined by potentiometer VR1 and capacitor C4. By adjusting potentiometer VR1, you can vary the time period. The output of IC2 is applied to either timer IC3 or optocoupler IC4 (MOC3021)

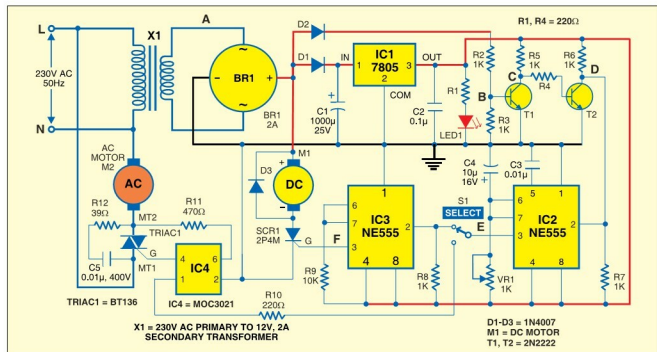


Fig. 1: Circuit for speed controller for AC and DC motors

